KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
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SDC1-WEEK1-Day2

II-I CSE, CSE(AI&ML) AY: 2026-2027

SL No	Week	Part	Detailed Subtopics	Part	Detailed Subtopics
1	Week 1	**Part A**	**Overfitting: Causes & Solutions, Regularization (L1 - Lasso, L2 - Ridge)**	Part B	Evaluation metrics

CONTENTS
(Week 1-Session 2 Part A: Overfitting and Regularization)

1. What Happens When We Increase the Degree?

Increasing the **degree** of a Polynomial Regression model allows it to fit more complex, curved relationships. The degree directly controls model complexity, which affects the risk of underfitting or overfitting:

- A **low degree** may not capture the pattern in the data → **Underfitting**
- A **very high degree** fits the training data almost perfectly, but performs poorly on new data → **Overfitting**

2. Visualizing Overfitting Across Polynomial Degrees

The plot below fits the **same dataset** with three different polynomial degrees, using a standard train/test scatter format (blue = training points, orange = testing points, black line = fitted curve).

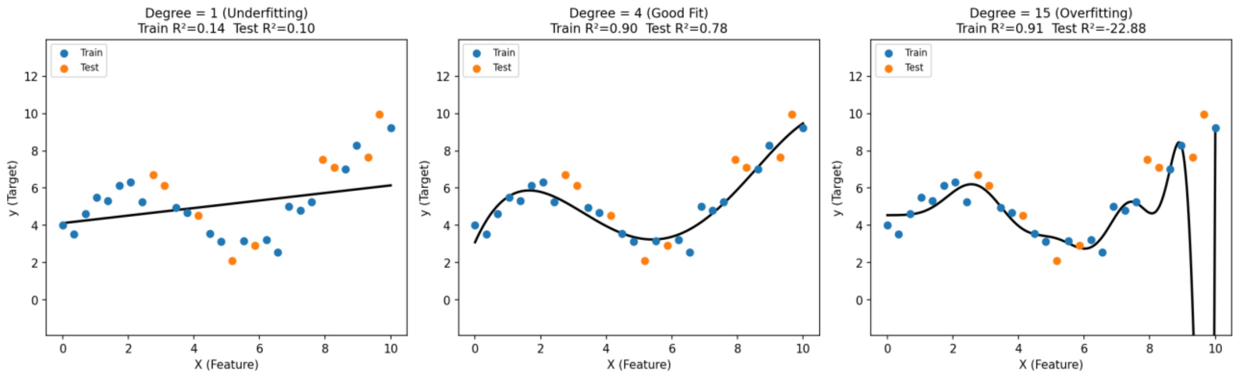

Figure: Effect of increasing polynomial degree on the fitted curve and Train/Test R^2

Degree = 1 (Left Panel) – Underfitting

The straight line is too simple to follow the curve in the data. Both **Train R^2 and Test R^2 are low** — the model has not learned the pattern in the training data.

Degree = 4 (Middle Panel) – Good Fit

The curve follows the general shape of the data without chasing every point. **Train R^2 and Test R^2 are both reasonably high and close to each other**, indicating a good balance between bias and variance.

Degree = 15 (Right Panel) – Overfitting

The curve bends sharply and oscillates wildly, especially near the edges of the data, in order to pass through every single training point. **Train R^2 stays high, but Test R^2 drops sharply** (even going negative) — the model has fitted the training noise instead of the true pattern.

This progression — straight line, smooth curve, wild oscillating curve — is the visual signature of underfitting, good fit, and overfitting as polynomial degree increases.

3. What Is Overfitting?

Overfitting happens when a model learns the training data too well, including its noise, instead of learning the actual underlying pattern.

Such a model gives very high accuracy on the training data but performs poorly on new, unseen data.

Signs of Overfitting

- Training R^2 / accuracy is very high, but testing R^2 / accuracy is much lower
- The fitted curve becomes overly wavy, bending to touch every data point
- Model weights become very large in magnitude

Real time Example:

Imagine a student preparing for an exam.

The student **memorizes all the practice questions and their answers** instead of understanding the concepts.

- **Practice test:** 100%
- **Actual exam with new questions:** 40%

The student performed extremely well on the questions they had already seen but poorly on new questions.

In Machine Learning:

- **Practice questions → Training data**
- **Memorizing questions → Learning training data too closely**
- **Practice test → Training performance**
- **New exam questions → Testing/unseen data**

4. What Is Underfitting?

Underfitting occurs when a model is too simple to learn the important patterns in the training data, resulting in poor performance on both training and unseen data.

Real time Example:

Imagine a student preparing for an exam.

The student studies **only a few basic concepts** and does not practice enough questions.

- **Practice test:** 50%
- **Actual exam:** 45%

The student performs poorly on **both familiar and new questions** because they haven't learned enough.

In Machine Learning:

- **Practice questions → Training data**
- **Studying too little → Model is too simple**
- **Practice test → Training performance**
- **Actual exam → Testing performance**

5. Bias and Variance: The Underlying Cause of Overfitting and Underfitting

The concepts of **bias** and **variance** explain **why** overfitting and underfitting occur. Every model's prediction error comes from a combination of these two sources.

Bias

Bias is the error caused by a model being **too simple** to capture the real pattern in the data. A high-bias model makes strong, oversimplified assumptions — for example, assuming a straight-line relationship when the true relationship is curved.

- High bias → the model is **too rigid** → it **underfits** — poor performance on both training and testing data

Variance

Variance is the error caused by a model being **too sensitive** to the specific training data it saw, including its noise and random fluctuations. A high-variance model changes drastically if the training data changes slightly.

- High variance → the model is **too flexible** → it **overfits** — excellent performance on training data, poor performance on testing data

Overfitting and Underfitting, Defined Through Bias and Variance

	Underfitting	**Overfitting**
Bias	High — model is too simple	Low — model is flexible enough
Variance	Low — model barely reacts to data	High — model reacts to every fluctuation
Train Error	High	Very Low
Test Error	High	High
Polynomial Degree Example	Degree = 1 (too simple)	Degree = 15 (too flexible)

In summary: **Underfitting = High Bias, Low Variance**. **Overfitting = Low Bias, High Variance**. A good model finds the middle ground — low enough bias to learn the real pattern, and low enough variance to ignore the noise. This balance is called the **Bias–Variance Tradeoff**.

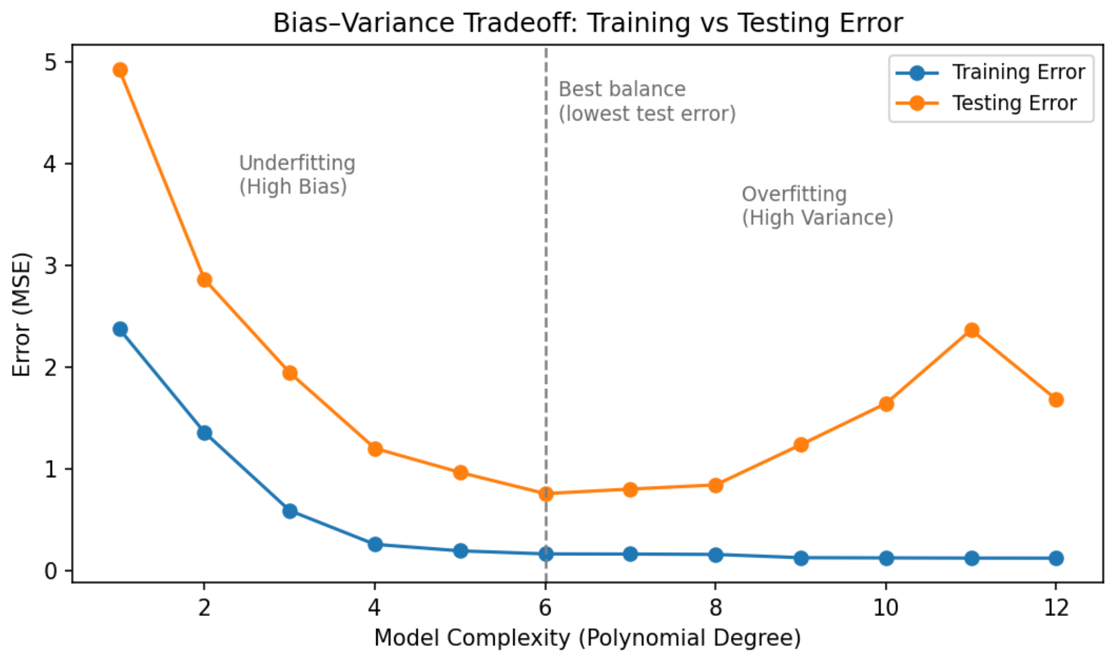

Figure: As model complexity (degree) increases, training error keeps falling, but testing error falls and then rises again — the lowest point on the testing curve is the best bias–variance balance.

6. Why Does Overfitting Happen?

Overfitting mainly happens because of:

- **Too high model complexity:** Using a very high polynomial degree gives the model too much freedom to fit every point exactly.
- **Too little training data:** With few data points, a complex model can easily memorize them instead of learning the true trend.
- **No control on weights:** Nothing stops the model from assigning very large weights to higher-degree terms, causing sharp, unnatural bends in the curve.

Example: a degree-10 Polynomial Regression model may show Train $R^2 \approx 0.99$ but Test R^2 far lower — a clear sign it has memorized the training data instead of learning the real relationship, exactly as seen in the degree = 15 plot above.

7. How Do We Detect Overfitting?

Overfitting is detected by **comparing performance on training data with performance on testing data.** A large gap between the two — training score much higher than testing score — indicates overfitting.

- Compute R^2 / MSE separately on the training set and the testing set
- Plot the fitted curve — an overfit curve looks unnaturally wavy, as seen in the degree = 15 plot above

8. Solutions to Overfitting

Several strategies exist to reduce overfitting. The complete set of solutions is listed below:

1. **Reduce Model Complexity:** Lower the polynomial degree, or choose a simpler model, so it has less freedom to chase noise.
2. **Get More Training Data:** More examples make it much harder for the model to simply memorize individual points.
3. **Cross-Validation:** Use k-fold cross-validation to choose the degree / model that generalizes best, instead of the one with the best training score.
4. **Feature Selection:** Remove irrelevant, redundant, or highly correlated features/terms before training.
5. **Early Stopping:** Stop training once validation performance stops improving (mainly used for iterative learners such as gradient descent and neural networks).
6. **Regularization (L1 / L2):** Add a penalty term to the cost function that discourages large weights.
7. **Pruning:** For tree-based models, cut back branches that fit noise rather than genuine patterns.
8. **Dropout:** For neural networks, randomly deactivate some neurons during training so the network cannot over-rely on any single one.
9. **Ensembling:** Combine multiple models (e.g., bagging, random forests) so their individual overfitting tendencies average out.

9. Regularization: The Core Idea

A high-degree Polynomial Regression model overfits because it is free to assign any value, however large, to its weights (w_1, w_2, …, w_n), so as to pass through every training point.

Regularization controls this by adding a **penalty term** to the cost function that discourages large weight values. The model must now balance two things:

1. Fit the training data well, and
2. Keep the weights small.

The plain regression cost function (Mean Squared Error) is:

$$\mathbf{MSE = (1/n) \times \Sigma\, (y_i - \hat{y}_i)^2}$$

n = number of data points
yi = actual (observed) value
ŷi = value predicted by the model

Regularization adds a penalty multiplied by a parameter **λ (lambda)**, called **alpha** in scikit-learn. Increasing alpha increases the penalty, pushing weights smaller and reducing variance (at the cost of some bias). The two common types are **L2 (Ridge)** and **L1 (Lasso)** regularization.

10. L2 Regularization – Ridge Regression

Ridge Regression adds a penalty equal to the sum of the squares of the weights:

$$\textbf{Cost} = \textbf{MSE} + \lambda \times \Sigma (w_j)^2$$

λ (lambda) = regularization strength (alpha in scikit-learn)
wj = the weight of the j-th feature

Because the penalty is based on **squared** values, larger weights are penalized much more heavily than smaller ones. This pushes the model to spread influence more evenly across features rather than letting one or two weights grow very large.

Effect of Ridge (L2)

- Shrinks all weights toward zero, but rarely makes them exactly zero
- Keeps all features/terms in the model, just with reduced influence
- Gives a smoother curve, less sensitive to individual data points
- Works well when most or all features are genuinely useful, especially when they are correlated with each other

How Alpha Affects Ridge

- alpha = 0 → same as plain regression (no regularization)
- Small alpha → mild shrinkage
- Large alpha → weights pushed close to zero → risk of underfitting

11. L1 Regularization – Lasso Regression

Lasso Regression (Least Absolute Shrinkage and Selection Operator) adds a penalty equal to the sum of the absolute values of the weights:

$$\textbf{Cost} = \textbf{MSE} + \lambda \times \Sigma |w_j|$$

λ (lambda) = regularization strength (alpha in scikit-learn)
wj = the weight of the j-th feature

Because the penalty grows **linearly** with weight size (rather than quadratically, as in Ridge), Lasso can push the **least useful** weights all the way down to **exactly zero**, effectively removing those terms from the model entirely.

Effect of Lasso (L1)

- Can shrink some weights exactly to zero
- Effectively removes unimportant terms — performs automatic feature selection
- Gives a simpler, sparser model than Ridge
- Especially useful when only a few features/terms are expected to matter

How Alpha Affects Lasso

- alpha = 0 → same as plain regression
- Small alpha → a few, mildly useful terms may be zeroed out
- Large alpha → many weights become zero → risk of underfitting (an overly sparse model)

12. Ridge vs Lasso – Detailed Comparison

Feature	**Ridge (L2)**	**Lasso (L1)**
Penalty Term	Sum of squared weights	Sum of absolute weights
Weight Shrinkage	Shrinks toward zero(Shrinkage)	Can become exactly zero (Shrinkage + Feature Selection)
Feature Selection	No — keeps all features	Yes — drops irrelevant features
Resulting Model	Dense (all terms retained, small weights)	Sparse (some terms removed)
Handles Correlated Features	Distributes weight smoothly among them	May arbitrarily keep one, zero out the rest
Best Suited For	Most / all features are useful	Only a few features truly matter
scikit-learn Class	Ridge(alpha=...)	Lasso(alpha=...)

Note: A combination of both penalties is called **Elastic Net**, which balances Ridge's smooth shrinkage with Lasso's feature-selection ability.

13. Advantages and Disadvantages

Advantages

- Reduces overfitting and improves performance on unseen data
- Lasso performs automatic feature selection
- Makes the model less sensitive to noise in the training data

Disadvantages

- The regularization strength (alpha) must be carefully chosen
- Too much regularization can cause underfitting
- Features usually need to be scaled before applying Ridge or Lasso

14. Applications

- **Healthcare:** Predicting disease severity from correlated clinical or voice features (e.g., Parkinson's severity scoring); Ridge preserves signal across all measurements.

- **Finance:** Stock price and credit-risk models with many correlated indicators, where regularization stabilizes weights.
- **Salary / Price Prediction:** Smoothing polynomial models on small datasets so the curve reflects the real trend, not noise.
- **Genomics:** Gene-expression studies with thousands of genes but few samples; Lasso selects the genes most relevant to an outcome.
- **Marketing Analytics:** Predicting customer lifetime value or churn from overlapping behavioural features, avoiding over-reliance on any single signal.
- **Text and Recommendation Systems:** High-dimensional inputs (text features, user-item data), where regularization keeps models generalizable.

Q & A

1. What is overfitting?

Answer:
Overfitting occurs when a model learns the noise in the training data along with the actual pattern, resulting in very high training accuracy but poor testing accuracy.

2. Why does overfitting happen in Polynomial Regression?

Answer:
It happens when the polynomial degree is too high, giving the model excessive freedom to fit every training point closely, including random noise, instead of learning the true relationship.

3. How can you tell if a model is overfitting?

Answer:
By comparing training and testing performance. If training R^2 is very high but testing R^2 is much lower, the model is overfitting.

4. What is regularization, and how does it reduce overfitting?

Answer:
Regularization adds a penalty term to the cost function based on the size of the model's weights. This discourages very large weights, keeping the model simpler and less likely to overfit.

5. What is the difference between L1 (Lasso) and L2 (Ridge) regularization?

Answer:
Ridge (L2) penalizes the sum of squared weights and shrinks them toward zero without eliminating them. Lasso (L1) penalizes the sum of absolute weights and can shrink some of them exactly to zero, effectively performing feature selection.

Scenario-Based Questions

6. Scenario-Based Question

Scenario:
A degree-10 Polynomial Regression model shows Train R^2 = 0.99 and Test R^2 = 0.70.

Question:
What is happening, and how would you fix it?

Answer:
The model is overfitting — it has memorized the training data instead of learning the true trend. This can be fixed by reducing the polynomial degree or applying Ridge or Lasso regularization.

7. Scenario-Based Question

Scenario:
After applying Ridge Regression, the test R^2 improves and the fitted curve becomes smoother.

Question:
Why does this happen?

Answer:
Ridge regularization shrinks large weights, reducing the influence of higher-degree terms that were causing sharp, unnatural bends in the curve. This gives a smoother curve that generalizes better to test data.

8. Scenario-Based Question

Scenario:
After applying Lasso Regression, several weights become exactly zero.

Question:
What does this mean?

Answer:
It means Lasso has removed the polynomial terms that were not contributing meaningfully to the prediction, giving a simpler model — this is Lasso's automatic feature selection.

9. Scenario-Based Question

Scenario:
A student says: “Regularization always improves a model.”

Question:
Do you agree? Why or why not?

Answer:
Not entirely. If the regularization strength (alpha) is set too high, the model can be penalized too much, leading to underfitting instead. Alpha must be chosen carefully, usually through cross-validation.

10. Scenario-Based Question

Scenario:
A student trains three models on the same dataset and records the following results:
Model A — Degree = 1: Train $R^2 = 0.32$, Test $R^2 = 0.29$
Model B — Degree = 12 (no regularization): Train $R^2 = 0.99$, Test $R^2 = 0.41$
Model C — Degree = 12 with Ridge (alpha = 5): Train $R^2 = 0.93$, Test $R^2 = 0.88$

Question:
Explain what is happening in each model in terms of bias and variance, and state which model you would deploy.

Answer:
Model A has high bias and low variance — it is too simple, so both its train and test scores are low (underfitting). Model B has low bias but very high variance — it fits the training data almost perfectly but fails on test data, shown by the large gap between Train R^2 (0.99) and Test R^2 (0.41), which is overfitting. Model C uses the same degree = 12 as Model B, but Ridge regularization reduces variance by shrinking the large weights — the training score drops slightly to 0.93, but the test score rises sharply to 0.88, showing far better generalization. Model C should be deployed: it gives up a small amount of training accuracy in exchange for a much better bias–variance balance, which is exactly the trade regularization is designed to make.